

# **Musculoskeletal Radiograph Abnormality Detection (MURA) Case Studies**

## **Evaluating ResNet and SWIN Transformer on the MURA Baseline**

**Shankar Chitrakar**

### **1. ABSTRACT**

The goal of this study is the use deep learning techniques to identify anomalies in musculoskeletal radiographs. Over .17 billion people worldwide suffer from musculoskeletal disorders, and diagnostic tools could increase the access to healthcare in areas where radiologist availability is limited. The MURA (musculoskeletal radiography) dataset, a sizable publicly accessible resource with 14,863 studies from 12,173 individuals and 40,561 multi-view radiographic pictures across seven upper extremity study types (elbow, finger, forearm, hand, humerus, shoulder, and wrist), is used in this project. Board-certified radiologists have manually classified each study in MURA as normal or abnormal.

The project will examine ResNet and SWIN Transformers' capabilities and contrast them with cutting-edge models. The optimal model for musculoskeletal X-ray interpretation is suggested by this thorough comparative investigation. This guarantees that the chosen method is accurate and efficient while yet being understandable for medical professionals.

The goal of this project is to advance medical imaging technologies for automated abnormality detection in musculoskeletal radiographs. The model's performance will be assessed on the task of differentiating between normal and abnormal studies using metrics like Cohen's kappa, which was used in the official MURA competition.

### **2. INTRODUCTION**

Musculoskeletal disorders represent a significant global health burden, accounting for a large proportion of radiographic imaging in clinical practice. The MURA (Musculoskeletal Radiographs) dataset, released by the Stanford Machine Learning Group, provides a large-scale benchmark for abnormal detection in musculoskeletal X-ray images. It consists of over 40,000 labeled images from seven upper extremity study types: elbow, finger, forearm, hand, humerus, shoulder, and wrist.

This project is to train models based on ResNet and SWIN Transformers to detect abnormalities in musculoskeletal radiographs. Identify models which are effective in medical image analysis tasks due to their ability to learn spatial hierarchies and extract discriminative features from raw pixel data.

## 2.1 Problem Statement

The objective is binary classification given a radiographic study (consisting of one or more views), determine if the study is **normal** or **abnormal**.

Dataset is created by researchers at Stanford University to support the development of machine learning models for identifying issues in X-ray images. This dataset is very useful as it allows researchers to improve the diagnosis of musculoskeletal disorders (MSDs), which are a leading cause of disability worldwide, as highlighted by ([3] Rajpurkar et al.) and the World Health Organization

## 2.2 Dataset Overview: MURA

MURA is a publicly available radiography dataset that includes 40,561 multi-view pictures from 14,863 investigations.

Board-certified radiologists manually classify it as abnormal or normal. The following distribution was used to comprehend the problem of class imbalance:

- The elbow, finger, forearm, hand, humerus, shoulder, and wrist are the seven upper extremity regions covered by the data.
- The Humerus and Forearm have the smallest portions of the dataset (~1,272 and ~1,825 photos), making them more prone to overfitting, whereas the Wrist and Shoulder have the biggest portions (~9,752 and ~8,379 images).
- The official baseline is a 169-layer DenseNet with an AUROC of 0.929. ([3] Rajpurkar and others)

The images in the MURA dataset are high-resolution grayscale images saved in PNG format. Sizes vary, with some images larger than 512 x 512 pixels, reflecting the diversity of real-world medical settings, where X-rays differ in size and quality. However, these 13 diversities also creates challenges for machine learning models, which must learn to interpret images with varying dimensions and qualities. Each patient case includes multiple views, such as front and side views for a wrist, which is useful because some conditions may only be visible from certain angles. Insights from the work of Chen et al. [32] emphasize that successful machine learning models need to analyze these varied perspectives to make accurate predictions.

The MURA dataset uses binary classification, labeling each X-ray as either normal or abnormal, based on whether a musculoskeletal issue is present. These labels, provided by trained radiologists, are highly reliable. In some studies, researchers have added more specific labels, such as fractures or dislocations, allowing models to identify both the presence and type of issue (Rajpurkar et al.)

One challenge with the MURA dataset is the subtlety of many abnormalities. Small fractures or early joint problems require careful analysis of fine details, and even trained radiologists sometimes miss these signs, which makes the hard to learn the task for machine learning models.

Another challenge is the varied quality of the X-ray images. Differences in resolution, exposure, and equipment used in different clinical settings make it hard for models to perform consistently. To address this, images are often preprocessed, including resizing to a consistent size and adjusting pixel values to balance exposure levels (Health Data Science Journal

2.2.1 Data Distribution

The dataset is categorized into seven standard upper extremity study types:

- **Elbow**
- **Finger**
- **Forearm**
- **Hand**
- **Humerus**
- **Shoulder**

2.2.2 Image stats

I. Train Data Set:

- **Number of unique patients: 11084**
- **Total Number of Images: 33128**

II Validation Data Set:

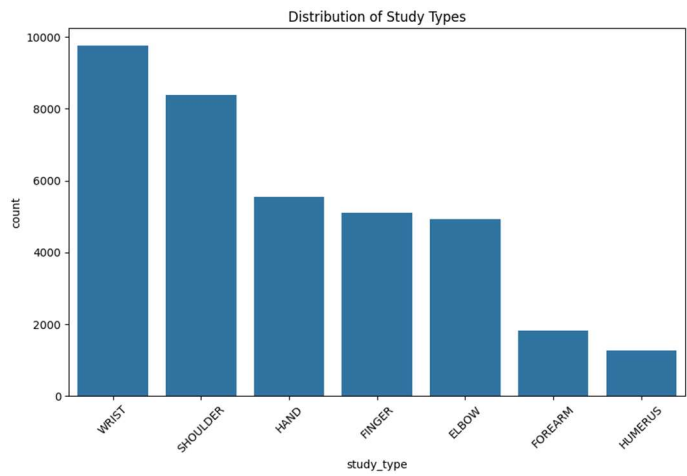
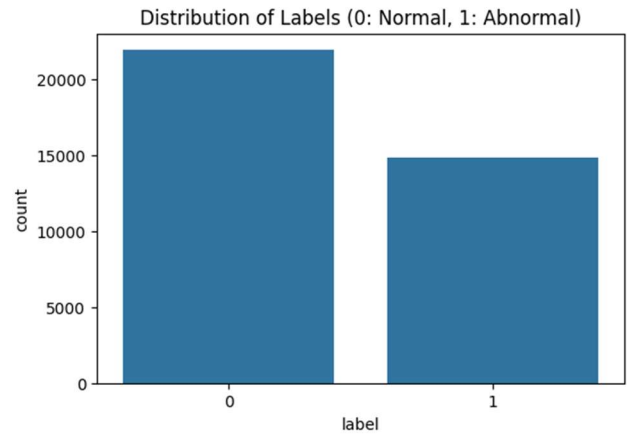
- **Number of unique patients: 3190**
- **Total Number of Images: 3680**

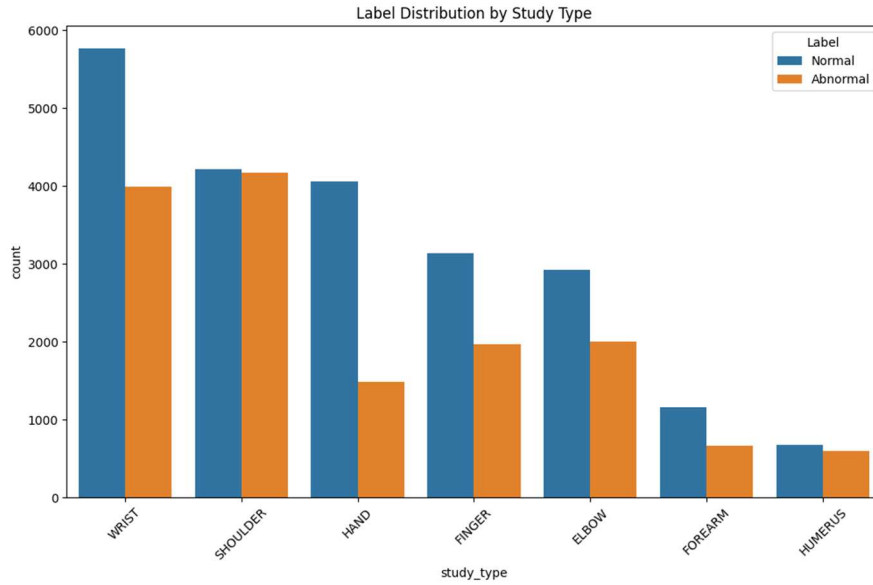
| Label negative positive |      |      | Label negative positive |     |     |
|-------------------------|------|------|-------------------------|-----|-----|
| Bone                    |      |      | Bone                    |     |     |
| XR_ELBOW                | 2644 | 1794 | XR_ELBOW                | 281 | 212 |
| XR_FINGER               | 2820 | 1775 | XR_FINGER               | 318 | 193 |
| XR_FOREARM              | 1059 | 584  | XR_FOREARM              | 105 | 77  |
| XR_HAND                 | 3653 | 1336 | XR_HAND                 | 406 | 148 |
| XR_HUMERUS              | 600  | 545  | XR_HUMERUS              | 73  | 54  |
| XR_SHOULDER             | 3775 | 3766 | XR_SHOULDER             | 436 | 402 |
| XR_WRIST                | 5159 | 3618 | XR_WRIST                | 606 | 369 |

Table 1: Summary of Training Image Data Set

Table 2: 1Summary of Validation Image Data Set

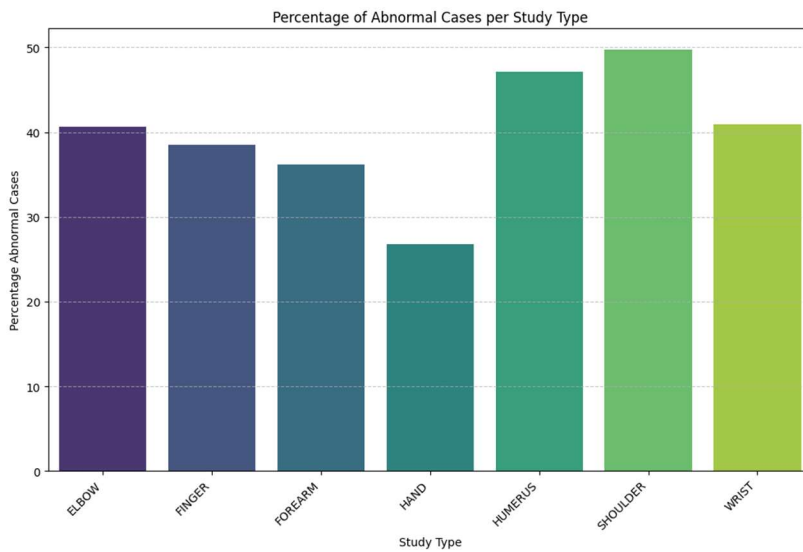
| Label    |       |  |  | Label    |      |  |  |
|----------|-------|--|--|----------|------|--|--|
| negative | 19710 |  |  | negative | 2225 |  |  |
| positive | 13418 |  |  | positive | 1455 |  |  |



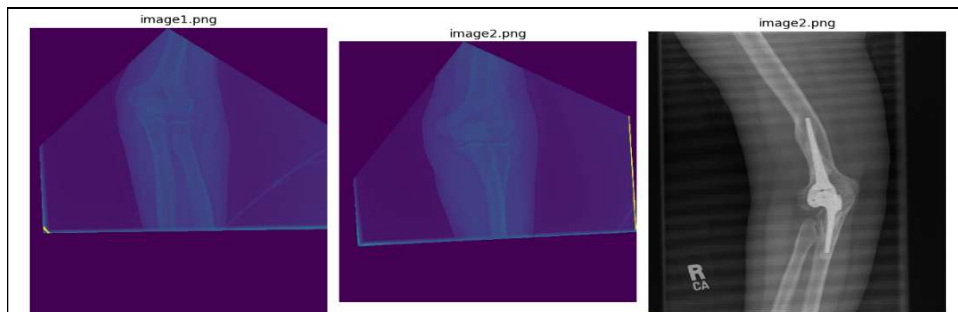


### 2.2.3 Percentage of Abnormal Cases per Study Type

| Study Type | Label 0 (Normal) | Label 1 (Abnormal) | Abnormal % |
|------------|------------------|--------------------|------------|
| Shoulder   | 0.502            | 0.497              | 49.74      |
| Humerus    | 0.529            | 0.47               | 47.09      |
| Wrist      | 0.591            | 0.408              | 40.88      |
| Elbow      | 0.593            | 0.406              | 40.68      |
| Finger     | 0.614            | 0.385              | 38.54      |
| Forearm    | 0.637            | 0.362              | 36.21      |
| Hand       | 0.732            | 0.267              | 26.77      |



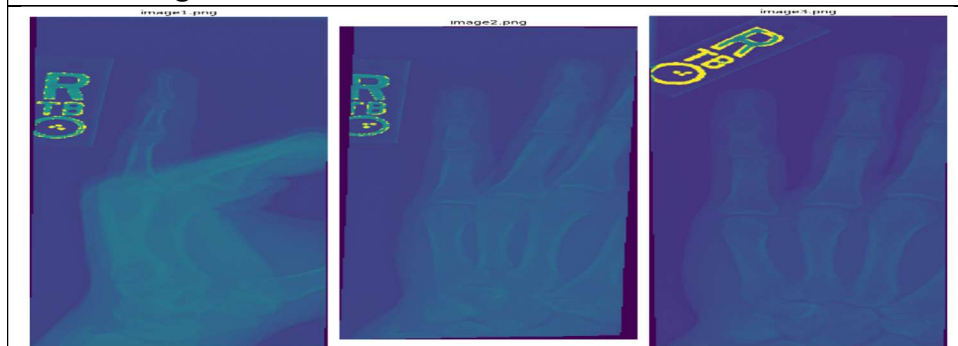
### 2.2.4 Sample Images



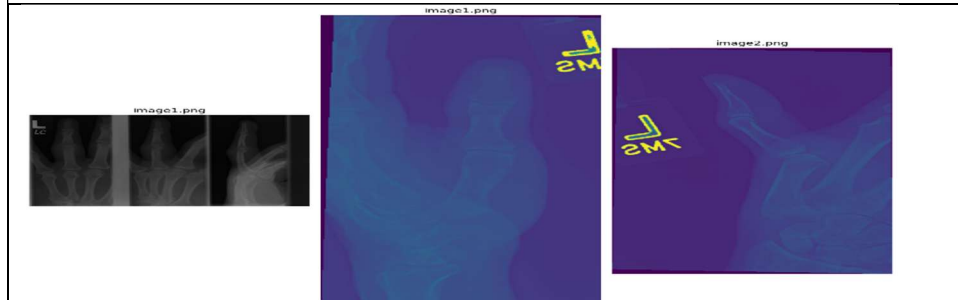
Picture: Positive Studies for XR\_ELLOW



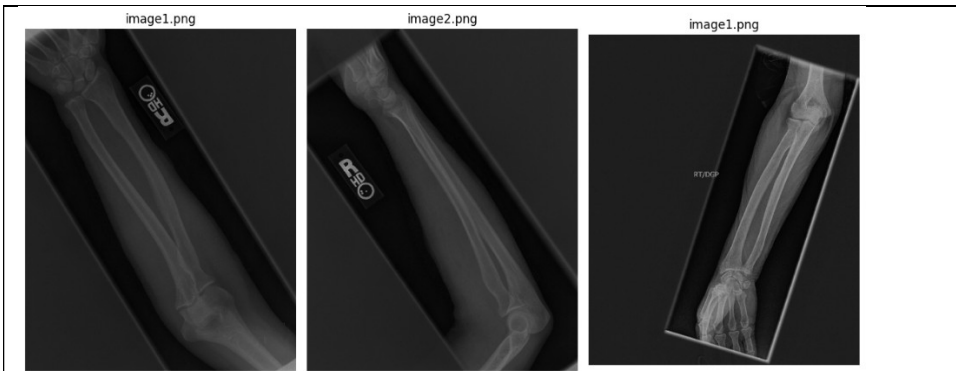
Picture: Negative Studies for XR\_ELLOW



Picture: Positive Studies for XR\_FINGER



Picture: Negative Studies for XR\_FINGER



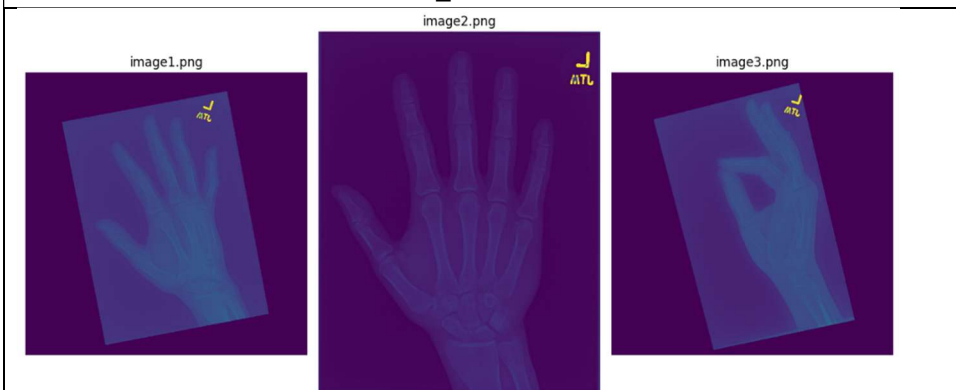
Picture: Positive Studies for XR\_FOREARM



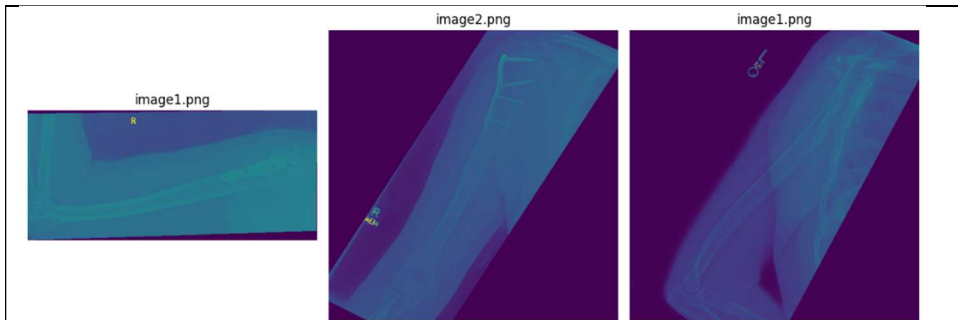
Picture: Negative Studies for XR\_FOREARM



Picture: Positive Studies for XR\_HAND



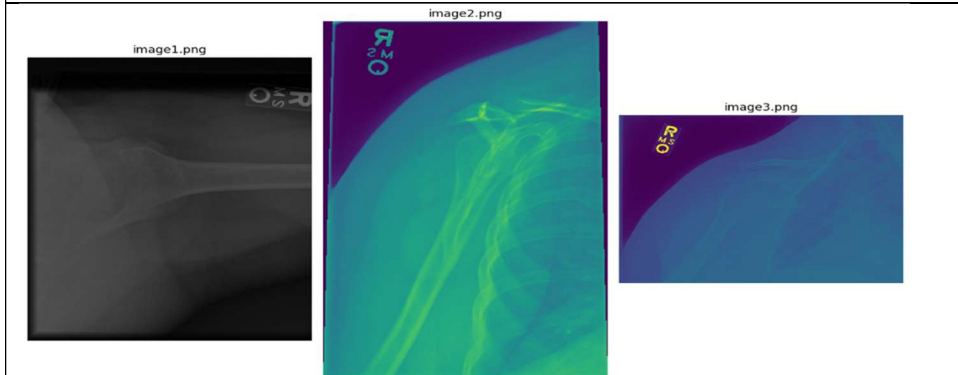
Picture: Negative Studies for XR\_HAND



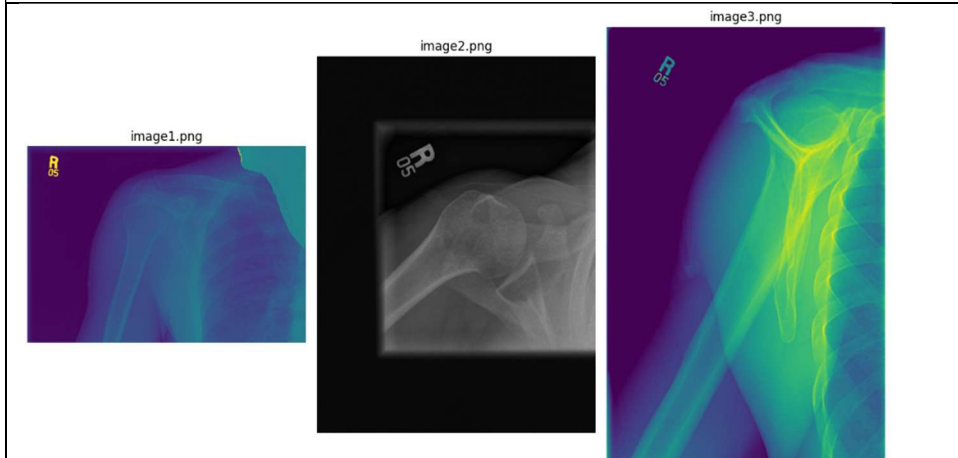
Picture: Positive Studies for XR\_HUMERUS



Picture: Negative Studies for XR\_HUMERUS

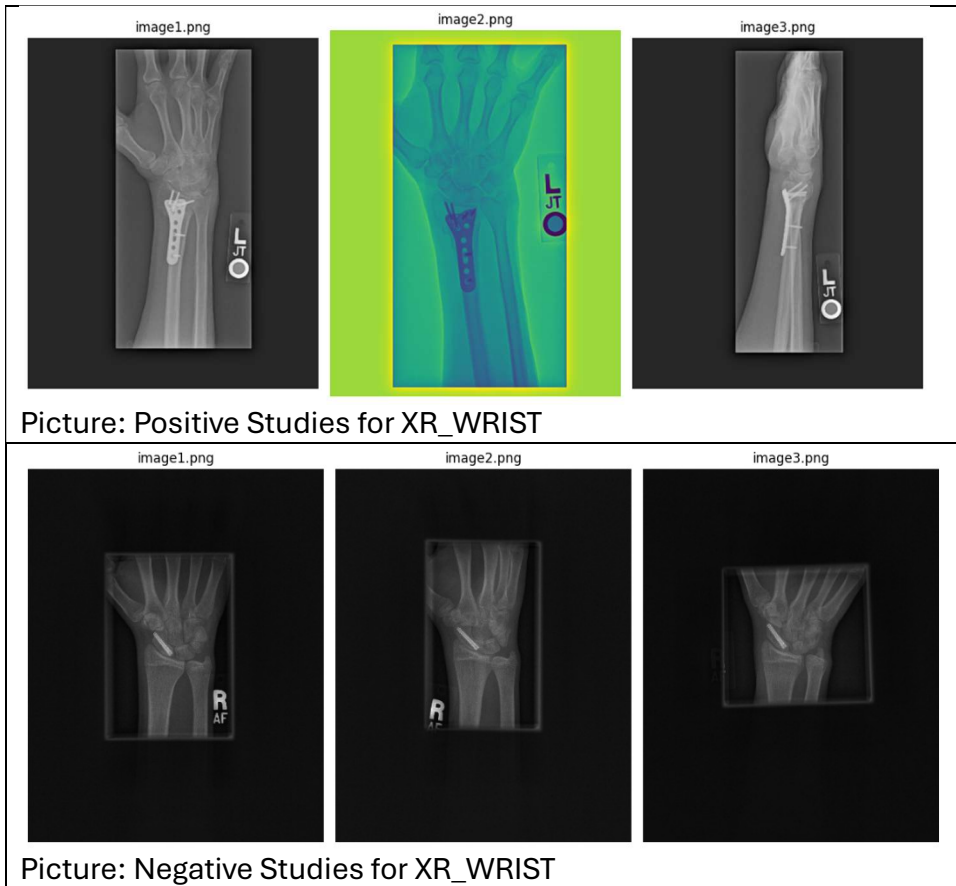


Picture: Positive Studies for XR\_SHOULDER



Picture: Negative Studies for XR\_SHOULDER





### 3. METHODOLOGY

This study will assess ResNet (18 and 50) and SWIN Transformer machine learning models to analyze the models using the MURA dataset, which includes a variety of X-ray images of different body parts. Each model's performance is evaluated based on accuracy, F1 scores, and computational efficiency, with an emphasis on capturing both fine details and more general patterns within medical images. Finding the optimal model for identifying minor musculoskeletal problems is the aim.

#### 3.1 ResNet (Residual Network) Architecture

ResNet18 was implemented to provide a robust convolutional baseline.

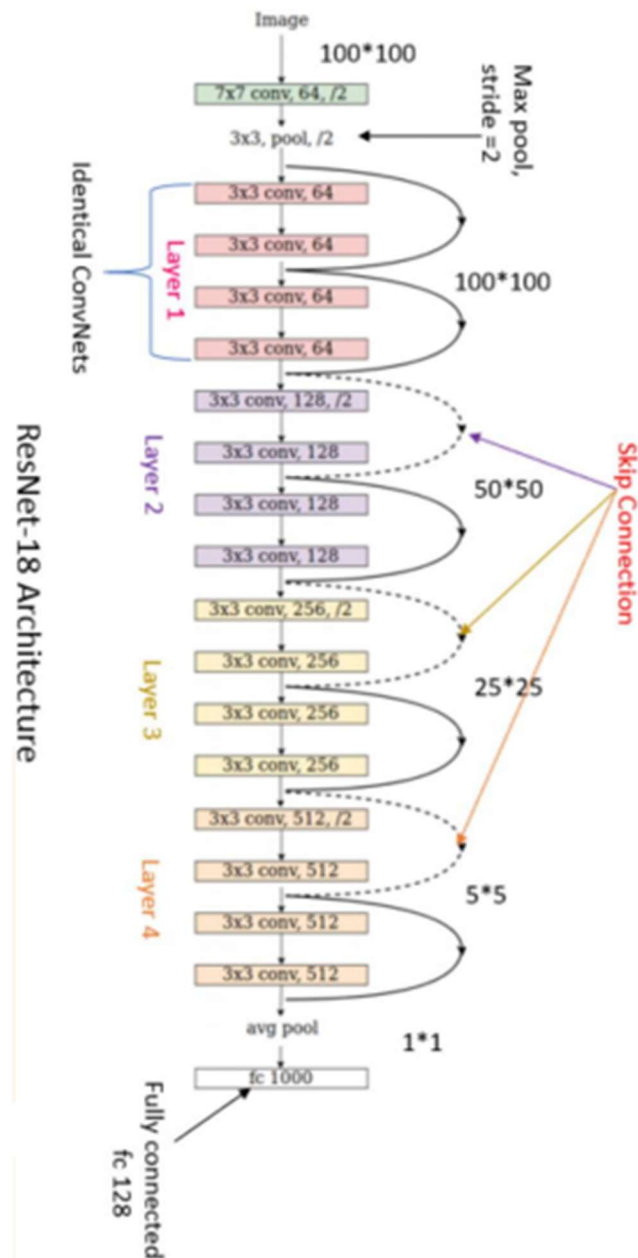
- Mechanism: It utilizes Residual Blocks with Skip Connections to allow gradients to flow through 18 layers without vanishing, which is essential for capturing the subtle textures of bone density and fractures in grayscale images.

- Modifications: The input layer was adjusted to accept single-channel X-rays, and the final fully connected layer was modified for binary classification (Normal vs. Abnormal).

### *1. The Core Innovation: Residual Learning*

The fundamental problem with very deep neural networks is the vanishing gradient, where the signal used to update weights becomes too small as it moves back through the layers.

- Skip Connections (Identity Mapping): ResNet introduces "shortcut" or "skip" connections that bypass one or more layers.
- Mathematical Principle: Instead of trying to learn a direct mapping  $H(x)$ , the network learns the residual mapping  $F(x) = H(x) - x$ .
- Benefit for MURA: This allows the model to be much deeper (e.g., ResNet-18, ResNet-50 or ResNet-101) than standard CNNs, enabling it to learn the complex, fine-grained features of bone fractures and hardware without losing information during training.



**Figure: ResNet-18 Base Architecture ([6] ResNet18)**

## 2. ResNet-18 Vs ResNet-50

### 2.1 Design

The most significant difference between ResNet-18 and ResNet-50 is the internal structure of the residual blocks.

- **ResNet-18 (Basic Block):** Uses two 3 X 3 convolutional layers.
- **ResNet-50 (Bottleneck Block):** Uses a three-layer stack:
  1. **1 X1 Conv:** Reduces the number of input channels (the "bottleneck").

2. **3 X 3 Conv:** Performs the main spatial feature extraction.
  3. **1 X1 Conv:** Restores the channel dimensions to match the output requirement.
- **Purpose:** This design allows the network to be much deeper (50 layers vs 18) without a massive explosion in the number of parameters, as the 3 X 3 convolution operates on a smaller "bottlenecked" dimension.

## 2.2. Layer Configuration and Stages

ResNet-50 follows the same 5-stage macro-architecture as ResNet-18, but with significantly increased density within each stage:

- **Stage 1 (Stem):** Identical to ResNet-18. A 7 X 7 convolution (stride 2) followed by 3 X 3 Max Pooling.
- **Stage 2 (conv2\_x):** Contains **3** Bottleneck blocks. Output depth is 256.
- **Stage 3 (conv3\_x):** Contains **4** Bottleneck blocks. Output depth is 512.
- **Stage 4 (conv4\_x):** Contains **6** Bottleneck blocks. Output depth is 1024.
- **Stage 5 (conv5\_x):** Contains **3** Bottleneck blocks. Output depth is **2048**.

Totaling these layers plus the initial convolution and final fully connected layer gives you the "50" in ResNet-50 ( $1 + 3 \times 3 + 4 \times 3 + 6 \times 3 + 3 \times 3 + 1 = 50$ ).

## 2.3. Impact on MURA Training

Switching from ResNet-18 to ResNet-50 affects your training pipeline in several ways:

- **Feature Complexity:** With a final feature map depth of **2,048** (compared to 512 in ResNet-18), the model has a much larger capacity to represent complex bone abnormalities, which can help improve your **ROC-AUC**.
- **Computational Cost:** ResNet-50 has roughly **23.5 million parameters**, which is more than double ResNet-18's ~11 million. This will result in slower training times per epoch and higher GPU memory usage.
- **Overfitting Risk:** Because ResNet-50 is much more "expressive," it is more prone to memorizing the training set. When using this model for MURA, it is critical to monitor your **Cohen's Kappa** on the validation set to ensure the model isn't just overfitting to the training distribution.

## 3. Implementation Specifics for MURA

To adapt ResNet for the MURA dataset, several specific modifications are typically made:

- Input Handling:
  - Resolution: Images are often resized to 224 X 224 or 512 X 512 to maintain consistency across the dataset.

- Preprocessing: Images are normalized using the mean and standard deviation of the ImageNet dataset.
- Study-Level Inference:
  - Because MURA is labeled per study (which may contain multiple images), the ResNet model processes each image in the study individually.
  - The final study-level prediction is the arithmetic mean of the probabilities calculated for each image.
- Binary Output: The final layer is modified to use a Sigmoid activation function, producing a probability between 0 (Normal) and 1 (Abnormal).
- Loss Function: Binary Cross-Entropy (BCE) is the standard loss used to train the network on the ground-truth radiologist labels.

#### 4. Comparison to the MURA Baseline

- While ResNet is a strong performer, the official MURA baseline was established using a 169-layer DenseNet (DenseNet-169).
- DenseNet Advantage: DenseNet connects every layer to every other layer within a block, which promotes feature reuse and is often slightly more efficient for small medical datasets than ResNet.
- ResNet Advantage: ResNet is generally faster to train and easier to implement, serving as the standard "modern" baseline for most musculoskeletal imaging research.

### 3.2 SWIN Architecture

The **SWIN Transformer** (Shifted Window Transformer) represents a paradigm shift in medical imaging by applying the attention mechanisms of natural language processing to visual data in a way that is computationally efficient and spatially aware.

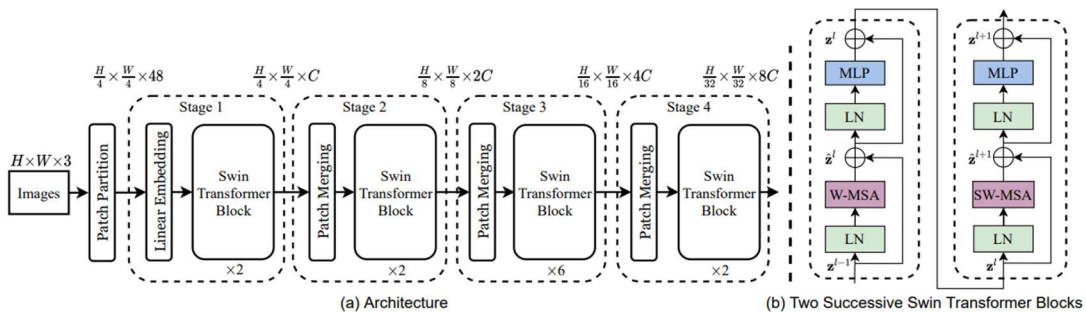


Figure: (a) The architecture of a SWIN Transformer (SWIN-T); (b) two successive SWIN Transformer Blocks. W-MSA and SW-MSA are multi-head self attention modules with regular and shifted windowing configurations, respectively. ([5] Liu et al., 2021)

The SWIN Transformer was implemented to evaluate the efficacy of attention-based models in a medical context.

**Mechanism:** It uses Shifted Window Self-Attention (SW-MSA) to provide a hierarchical representation of the image. This allows the model to "attend" to both local anomalies (like a hairline fracture) and global structures (like joint alignment).

**Efficiency:** Unlike standard transformers, SWIN has linear computational complexity relative to the image size, making it highly efficient for the high-resolution radiographs found in MURA.

### 1. Hierarchical Feature Maps

The SWIN Transformer builds **hierarchical feature maps**.

- **Patch Partitioning:** The input image (e.g., a 224 X 224 X-ray) is divided into small, non-overlapping patches (typically 4 X 4).
- **Linear Embedding:** These patches are projected into a 1D vector representing the initial "tokens" for the Transformer.
- **Merging Layers:** As the network progresses through its four stages, it merges neighboring patches. This reduces the spatial resolution (down sampling) while increasing the depth of the features, similar to the behavior of a CNN like ResNet.

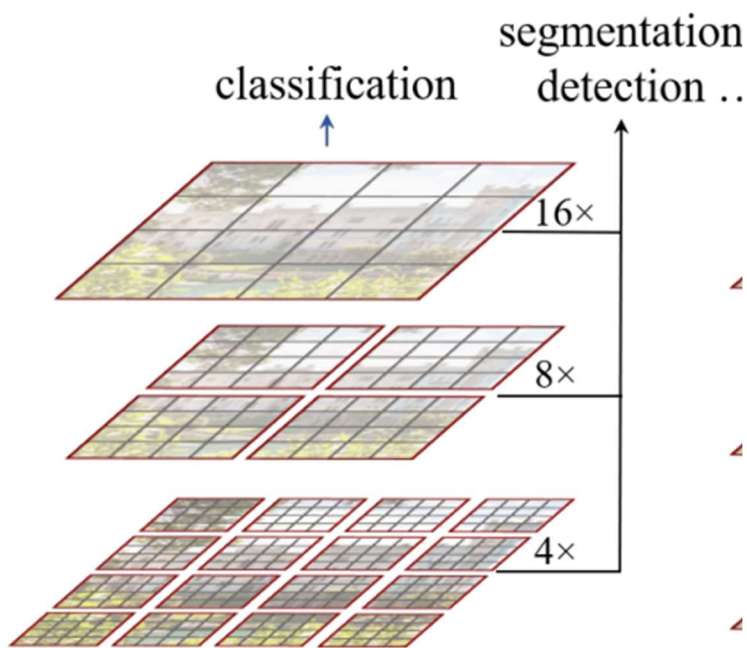


Figure: SWIN Transformer builds hierarchical feature maps by merging image patches (shown in gray) in deeper layers and has linear computation complexity to input image size due to computation of self-attention only within each local window (shown in red). It can thus serve as a general-purpose backbone for both image classification and dense recognition tasks ([5] Liu et al., 2021)

## 2. Shifted Window-Based Self-Attention

The most critical innovation of the SWIN architecture is how it handles "Attention" to remain efficient on high-resolution MURA images.

- **Local Window Attention:** Instead of calculating attention across the entire image (which is computationally expensive), SWIN calculates self-attention only within local, non-overlapping **windows** (e.g., 7 X 7 patches).
- **The Shifted Window (S-WSA):** To allow information to travel between separate windows, the partition is "shifted" between consecutive layers. This shift bridges the boundaries of the previous windows, providing the model with a broader context without increasing the math load.
- **Relevance to MURA:** This is vital for X-rays because a fracture (local detail) must be understood in the context of the entire joint alignment (global context).

## 3. SWIN-Tiny (SWIN-T) Configuration

For the MURA dataset, the **SWIN-Tiny** variant is frequently used as a starting point. Its architecture typically includes:

- **Stages:** 4 stages with varying numbers of Transformer blocks (e.g., 2, 2, 6, 2).
- **Attention Heads:** The number of heads increases at each stage (e.g., 3, 6, 12, 24), allowing the model to focus on multiple different features simultaneously.
- **Feature Dim:** The base dimension is usually 96.

## 4. Training Specifics for MURA

- **Softmax/Sigmoid Head:** The final layer is replaced with a linear head and a Sigmoid activation to output a probability of abnormality.
- **Study-Level Inference:** Similar to the ResNet and DenseNet implementations, the SWIN model predicts a score for every image in a study. These are then averaged to determine the final diagnosis for that musculoskeletal study.
- **Performance:** SWIN Transformers have demonstrated superior performance over ResNet on MURA, often achieving higher **AUROC** and **Cohen's Kappa** scores due to their superior ability to capture long-range dependencies in the bone structure.

| Feature                | ResNet (CNN)                      | SWIN Transformer                     |
|------------------------|-----------------------------------|--------------------------------------|
| <b>Basic Unit</b>      | Convolutional Kernel              | Shifted Window Attention             |
| <b>Spatial Bias</b>    | Strong local inductive bias       | Learns spatial relations dynamically |
| <b>Receptive Field</b> | Fixed (based on kernel size)      | Flexible (window-based)              |
| <b>MURA Advantage</b>  | Fast, stable, excellent for edges | Superior global/contextual reasoning |

## 4 Data Preprocessing

### 4.1 Data set:

Images are available in different shapes and sizes.

Requires resizing, transforming and normalization to tensor before training the model.

```
self.transform = transforms.Compose([
    transforms.Resize((512, 512)), # Resize to a common size
    transforms.ToTensor(),
    # Add this line to ensure 3 channels
    transforms.Lambda(lambda x: x.repeat(3, 1, 1) if x.shape[0] == 1 else x)
])
```

### 4.2. The Baseline: 169-layer DenseNet

The original baseline for MURA utilized a **169-layer DenseNet (DenseNet-169)**.

#### Architecture and Methodology

- **Per-View Prediction:** The model predicts the probability of abnormality for each individual image (view) in a study.
- **Study-Level Aggregation:** The final prediction for a study is calculated by taking the **arithmetic mean** of the probabilities across all its views.
- **Decision Threshold:** A study is classified as abnormal if the aggregated probability is greater than **0.5**.

### 4.3. Performance

The baseline model achieved an **AUROC of 0.929**. It performed comparably to board-certified radiologists on finger and wrist studies but lagged on more complex areas like the shoulder and humerus.



## 5. Implementation Architectures

### 5.1. ResNet (Residual Network)

ResNet is a foundational CNN architecture known for its use of **residual blocks** and **skip connections**, which allow for training very deep networks by mitigating the vanishing gradient problem.

- **Feature Extraction:** Uses localized convolutional kernels to identify spatial hierarchies.
- **Medical Suitability:** Effective at capturing local textures like bone density variations or fine fractures.

### 5.2. SWIN Transformer (Shifted Window)

The SWIN Transformer represents a shift from CNNs to attention-based models. Unlike the standard Vision Transformer (ViT), the SWIN architecture uses a **hierarchical structure** and **localized window-based attention**.

- **Multi-Scale Features:** It builds feature maps by gradually merging adjacent small patches, capturing both global context and fine-grained details.
  - **Efficiency:** The "shifted window" mechanism allows for cross-window communication without the massive computational cost of global self-attention.
- 

## 6. Methodology and Evaluation

### 6.1. Evaluation Methodology

**Primary Metric:** Following competition standards, **Cohen's Kappa** was used as the "gold standard" metric. This metric is superior to accuracy because it adjusts for the agreement between the AI and the radiologist that could occur purely by chance.

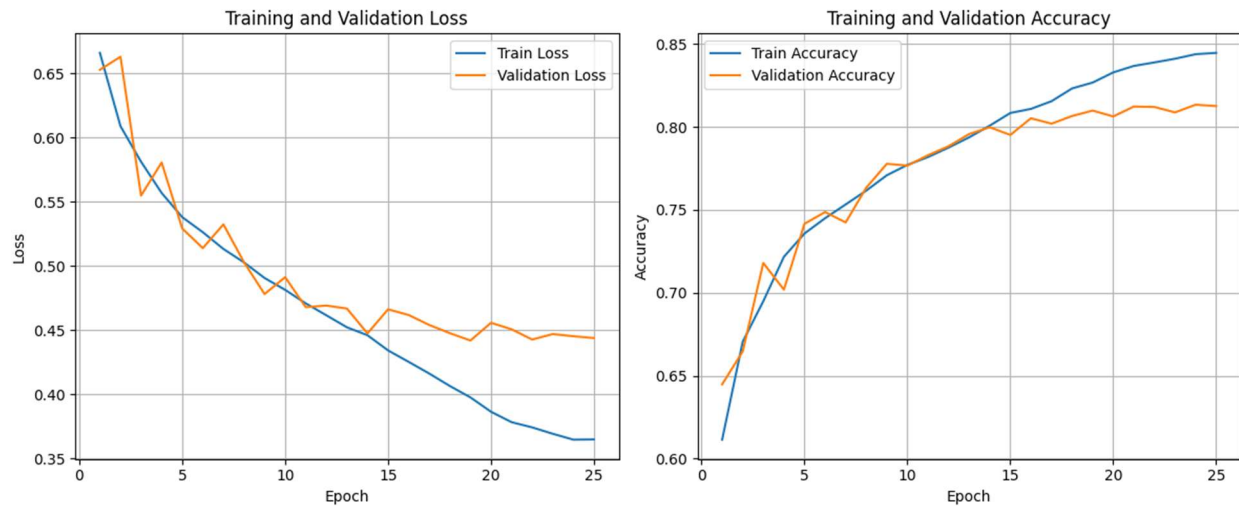
$$\kappa = \frac{p_o - p_e}{1 - p_e}$$

**Inference:** A **Study-Level** prediction was reached by calculating the **arithmetic mean** of the probabilities of all images within a single study.

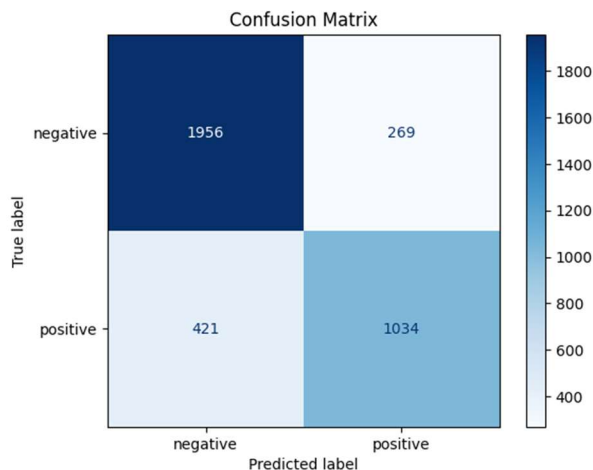
## 7 Results and Visualization of Experiments

### Resnet-18 Model:

#### Loss Vs Accuracy:

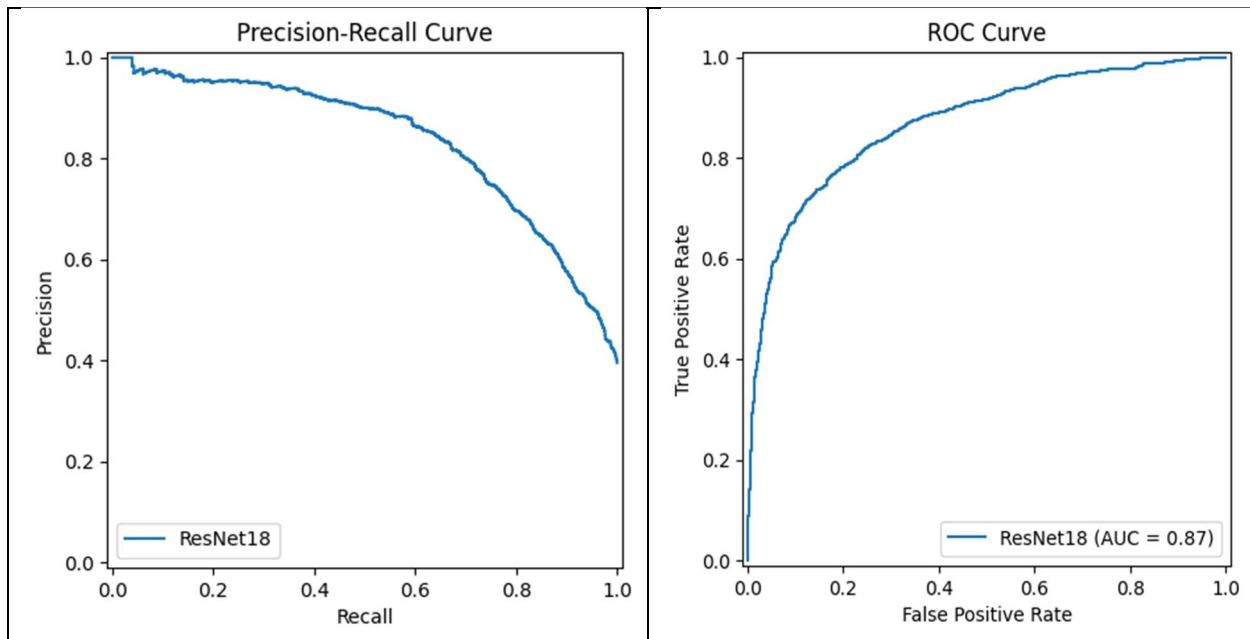


#### Confusion Matrix:



#### Precision-Recall Vs ROC

- Precision =  $TP / (TP + FP)$  [Of all case model claimed were abnormal, how many abnormal?]
- Recall =  $TP / (TP + FN)$  [Of all the *actual* abnormal cases in the dataset, how many did the model *correctly find*?]
- ROC: The ROC curve shows the tradeoff between the **True Positive Rate** and the **False Positive Rate**.



*Kappa*: 0.6006 (Substantial Agreement)

#### Additional Validations

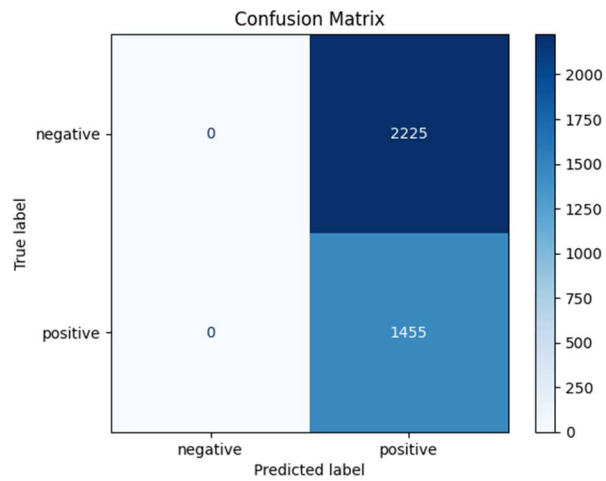
- Accuracy: 0.8125
- Precision: 0.7936
- Recall (Sensitivity): 0.7107
- F1-Score: 0.7498
- Specificity: 0.8791

#### Resnet-50 Model:

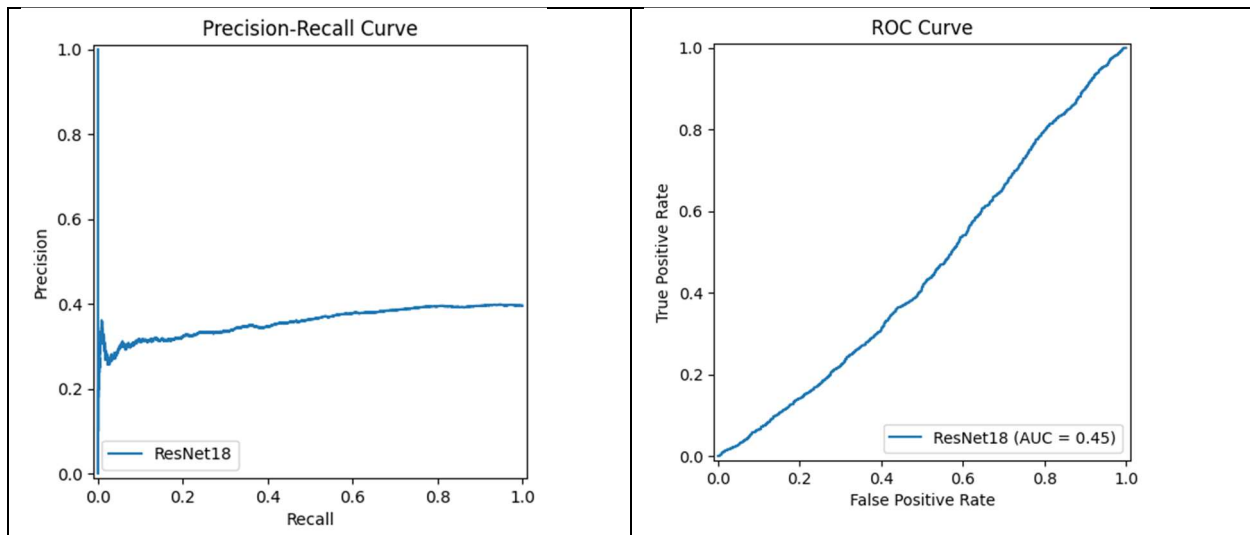
##### Loss Vs Accuracy:



### Confusion Matrix:



### Precision-Recall Vs ROC



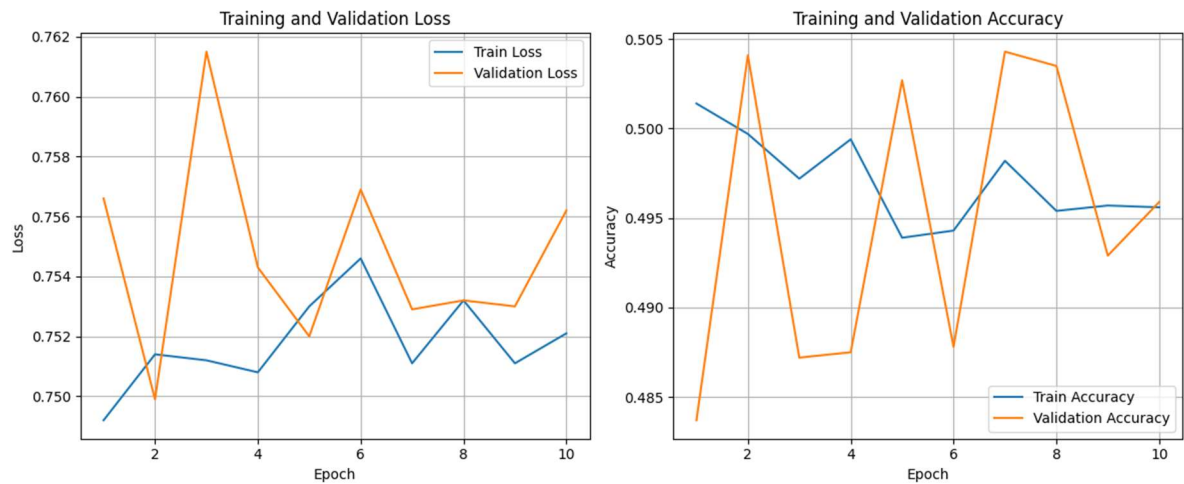
Cohen's Kappa: 0.0000

### Additional Validations

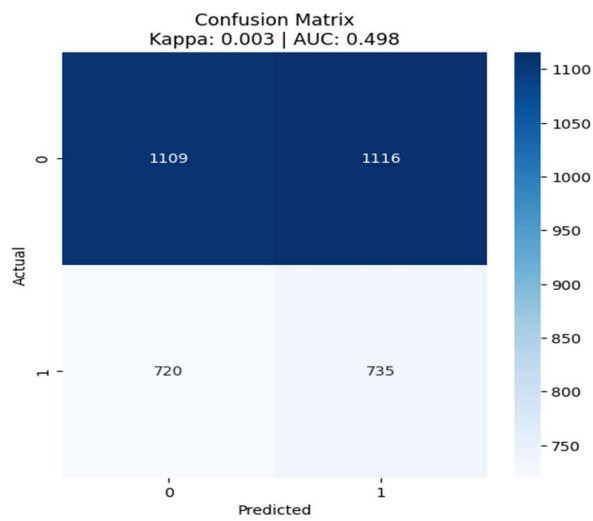
- Accuracy: 0.3954
- Precision: 0.3954
- Recall (Sensitivity): 1.0000
- F1-Score: 0.5667
- Specificity: 0.0000
- Cohen's Kappa: 0.0000

SWIN Transformer Results:

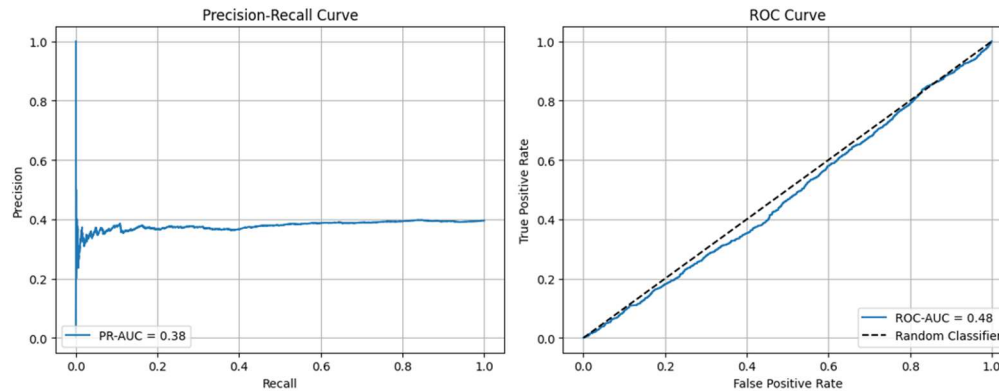
Loss Vs Accuracy:



Confusion Matrix:



## Precision-Recall Vs ROC



*Cohen's Kappa*: 0.0133 ( Slight agreement – almost random guessing)

## Additional Validations

- Overall Accuracy: 0.4832
- Precision per class (0, 1): [0.58810693 0.37899296]
- Recall (Sensitivity) per class (0, 1): [0.48449438 0.48109966]
- F1-score per class (0, 1): [0.53129621 0.42398546]

## Comparison Matrix

| Metric               | ResNet-18            | ResNet-50       | Swin Transformer        |
|----------------------|----------------------|-----------------|-------------------------|
| Accuracy             | 0.8125 (81.2%)       | 0.3954 (39.5%)  | 0.4832 (48.3%)          |
| Cohen's Kappa        | 0.6006 (Substantial) | 0.0000 (Random) | 0.0133 (Slight)         |
| Precision            | 0.7936               | 0.3954          | 0.3790 (Abnormal class) |
| Recall (Sensitivity) | 0.7107               | 1.0000*         | 0.4811 (Abnormal class) |
| Specificity          | 0.8791               | 0               | 0.4845 (Normal class)   |
| F1-Score             | 0.7498               | 0.5667          | 0.4240 (Abnormal class) |

## 7. Conclusion

This study shows the architectural complexity is not always guarantee the better performance in musculoskeletal abnormalities detection, in comparison with official benchmarks of 169-layer DenseNet ([3] Rajpurkar, Pranav, et al), ResNet-18 architecture provided the reliable clinical utility in agreement with radiologist labels. Whereas deeper Resne-50 and Swin Transformer showed optimization challenges, resulting in random performance. This indicates, model stability and proper convergence are as critical for improve healthcare access.

## 8. Final Conclusion

The study results show a contrast in performance: **ResNet-18** is the better model, achieving an accuracy of **81.25%** and a **Cohen's Kappa of 0.6006**.

Whereas the **ResNet-50** and **Swin Transformer** are failed to achieve better agreement (Kappa). This study shows the importance of architectural selection and the need for balanced precision and recall for AI tools that help radiologists in high-volume environments.

## 9. References:

[1] World Health Organization. "Musculoskeletal Conditions." World Health Organization, 2020. Available at:

<https://www.who.int/news-room/factsheets/detail/musculoskeletal-conditions>

[2]

<https://stanfordmlgroup.github.io/competitions/mura/#:~:text=MURA%20is%20a%20data set%20of,What%20is%20our%20baseline?>

[3] Rajpurkar, Pranav, et al. "MURA: Large Dataset for Abnormality Detection in Musculoskeletal Radiographs." arXiv preprint arXiv:1712.06957, 2017.

<https://arxiv.org/abs/1712.06957>

[4] Chen, Liyang, et al. "Dense Connectivity and Its Application in Medical Image Analysis." Biomedical Signal Processing and Control, vol. 68, 2021, p. 102709.4

[5] *SWIN Transformer: Hierarchical Vision Transformer using Shifted Windows* (Liu et al., 2021).

[https://arxiv.org/abs/2103.14030#:~:text=SWIN%20Transformer%20is%20a%20hierarchic al%20vision%20transformer,\\*%20\\*\\*Hierarchical%20Transformer\\*\\*%20\\*%20\\*\\*Shifted%20 windowing%20scheme\\*\\*](https://arxiv.org/abs/2103.14030#:~:text=SWIN%20Transformer%20is%20a%20hierarchic al%20vision%20transformer,*%20**Hierarchical%20Transformer**%20*%20**Shifted%20 windowing%20scheme**)

<https://arxiv.org/pdf/2103.14030>

[6] ResNet18

[https://www.researchgate.net/publication/364345322\\_Resnet18\\_Model\\_With\\_Sequential\\_Layer\\_For\\_Computing\\_Accuracy\\_On\\_Image\\_Classification\\_Dataset](https://www.researchgate.net/publication/364345322_Resnet18_Model_With_Sequential_Layer_For_Computing_Accuracy_On_Image_Classification_Dataset)